



Chanhee Park

M.S. candidate, Computer Science, Korea University · NLP&AI Lab

Research Interest : Benchmark Design, Agent Development and Evaluation, Domain Transfer, Inference Optimization

Research Achievement : 4 first-author papers (3 published, 1 arXiv), 2 Korean patent applications

Project Experience : 4 industry-academia projects, Tech lead for AI agent service in legal domain

*“I aim to diagnose **where** language models **fail**, and ship systems that **hold up in the field**.”*

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 [Website](#)

 [github](#)

 [LinkedIn](#)

Profile

Education

2014.02 Hanyoung Foreign Language High School
2020.02 B.A. Korean Language & Literature, Korea University
2027.02 M.S. Computer Science, Korea University (expected)

Projects

- **RAG System built for Breast-cancer Oncologists**
Built index on 1,356 ASCO guidelines for RAG pipeline
- **Sovereign Foundation Model (VAETKI) Development**
Built 6.5M SFT Q&A samples across 4 languages and 9 domains
- **LegalMi:dm Legal Vertical Model Training**
Led continual pre-training for Midm-2.0 (11.5B)
- **Korean Language Consultation Agents Development**
Built Evaluation metrics for KTC certification and evaluation harness

Career

2020.03–2023.06 Interpreting Officer, ROK Air Force
2023.06–2023.10 Goorm AI NLP Course (12th cohort)
2024.03–2025.03 Research Intern, NLP&AI Lab, Korea University
2025.03–2027.02 M.S., NLP&AI Lab, Korea University (adv. Heuseok. Lim)
2025.10– AI Engineer / Tech Lead, “아는노무사”

Publications

MIRAGE: A Metric-Intensive Benchmark for Retrieval-Augmented Generation Evaluation

C. Park, H. Moon, H. Lim, *NAACL Findings*, 2025

Can Small Open-Source Language Models With RAG Match GPT-4 Performance in Breast Cancer Clinical Decision Support?

C. Park, I.H. Park, M. Kim, Y. Jang, H. Lim, Y. Hur. *JCO Clinical Cancer Informatics*, 2026

Funnel-of-Thoughts: Efficient Test-Time Scaling via Early Voting and Rollout Pruning

C. Park, S. Han, J. Yoon, S. Hong, H. Lim. *ArXiv*, 2026

AgentHop: A Diagnostic Benchmark for Agentic Multi-Hop Scientific Question Answering

C. Park, J. Yoon, S. Han, H. Moon, H. Lim. *NeurIPS E&D*, 2026

Co-author on 2 international conference papers, 1 journal, 1 tech report, 1 workshop, 3 domestic, 2 preprints

MIRAGE: A Metric-Intensive Benchmark for Retrieval-Augmented Generation Evaluation

Noise Vulnerability

Query : When does the adventures of kid danger come out?
Answer : [January 19, 2018]
Doc_source : The Adventures of Kid Danger

Noise Span : The Adventures of Kid Danger is an American animated comedy television series created by Dan Schneider that aired on Nickelodeon from **January 15 to June 14, 2018**.

Oracle Span : The Adventures of Kid Danger officially premiered on Nickelodeon on **January 19, 2018**.

Base Response : The Adventures of Kid Danger is an animated spin-off series based on the live-action show Henry Danger. It premiered on Nickelodeon on **January 15, 2018**.

Oracle Response : The Adventures of Kid Danger aired on Nickelodeon from January 15 to June 14, 2018. It officially premiered on Nickelodeon on **January 19, 2018**.

Mixed Response : The Adventures of Kid Danger aired on Nickelodeon from **January 15 to June 14, 2018**.

Context Acceptability

Query : Who is the father of Sune Sik?
Answer : [Sverker I of Sweden]
Doc_source : Sune Sik

Oracle Span : Sune Sik, (born c. 1154), was allegedly a Swedish prince. According to Olaus Petri, he would have been a younger son of King **Sverker I of Sweden** and father of Ingrid Ylva.

Base Response : The father of Sune Sik is **Kim Sik**.

Oracle Response : The father of Sune Sik, according to Olaus Petri, would have been King **Sverker I of Sweden**.

Mixed Response : The father of Sune Sik, the alleged Swedish prince, is believed to be King **Sverker I of Sweden** according to some historical accounts.

Context Insensitivity

Query : What age group made up the highest percentage of the population?
Answer : [40 to 49]
Doc_source : Cunter

Oracle Span : Of the adult population, twenty-nine people or **14.6% of the population are between 20 and 29** years old. Twenty-eight people or 14.1% are 30 to 39, thirty-six people or **18.2% are 40 to 49**, and thirty-one people or 15.7% are 50 to 59.

Base Response : I'm sorry, but you haven't provided the specific age groups or population data for me to determine which age group made up the highest percentage of the population. Could you please provide more details so I can assist you better?

Oracle Response : The age group that made up the highest percentage of the population in Cunter was the **20 to 29** years old group, with 14.6% of the population falling within that age range.

Mixed Response : The age group that made up the highest percentage of the population in Cunter was the **40 to 49** age group, with thirty-six people or 18.2% of the population falling into that category.

Context Misinterpretation

Query : If the seaside resort in Victoria, Australia had retained its original name, what would Torquay be known as today?
Answer : [Spring Creek]
Doc_source : Torquay, Victoria

Oracle Span : From the 1860s, picnickers began to frequent the location, which was originally known as **Spring Creek**, after the watercourse along its south-western edge, but it was named **Puebla** in the 1882 Victorian Municipal Directory.

Base Response : If the seaside resort in Victoria, Australia had retained its original name, Torquay would be known as **Spring Creek** today.

Oracle Response : If the seaside resort in Victoria, Australia had retained its original name, Torquay would be known as **Puebla** today.

Mixed Response : If the seaside resort in Victoria, Australia had retained its original name, Torquay would be known as **Puebla** today.

Problem

When RAG is wrong, did retrieval return the wrong document, or did the generator fail to use it?

A single accuracy score cannot tell → no way to know what to fix

Method

Base: LLM alone / Oracle: gold document only / Mixed: top-5 incl. gold

	Base	Oracle	Mixed
Noise Vulnerability	-	O	X
Context Acceptability	-	O	O
Context Insensitivity	X	X	-
Context Misinterpretation	O	X	-

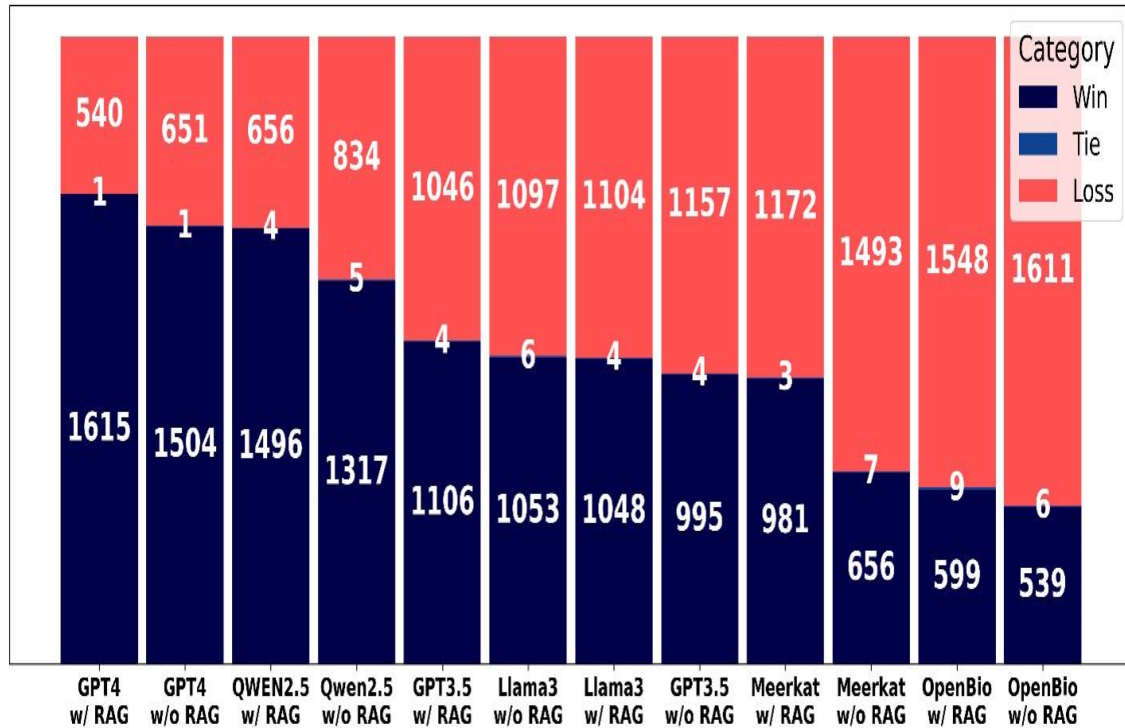
Data

5 public benchmarks → Wikipedia retrieval pool; LLM filtering + human review
7,560 queries · 37,800 documents; light enough to run locally

Results

Findings of NAACL 2025 · Korean patent filed · 1,700+ HuggingFace downloads
Methodology extended in two follow-up papers

Can Small Open-Source Language Models With RAG Match GPT-4 Performance in Breast Cancer Clinical Decision Support?



Problem

Medical LLMs are judged on multiple-choice benchmarks, but real practice demands long answers grounded in fast-changing guidelines → do benchmark scores predict clinical usefulness?

Method

- Joint project with Korea University Guro Hospital, Oncology
- Collected and chunked 1,356 ASCO breast-cancer guidelines
- SentenceTransformers + FAISS retrieval
- Evaluated on 98 queries written by clinicians
- 6 models (GPT-4 · GPT-3.5 · Qwen2.5 · Llama3-8B · Meerkat · OpenBioLLM)
- 12 generation settings (with / without RAG)
- 2,156 pairwise win/loss comparisons × 12 settings (LLM-as-a-judge)

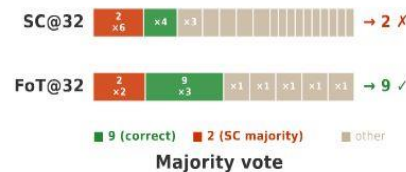
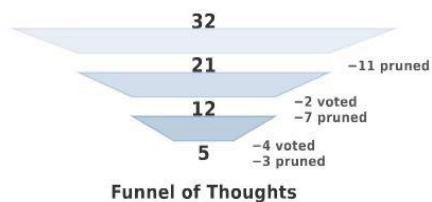
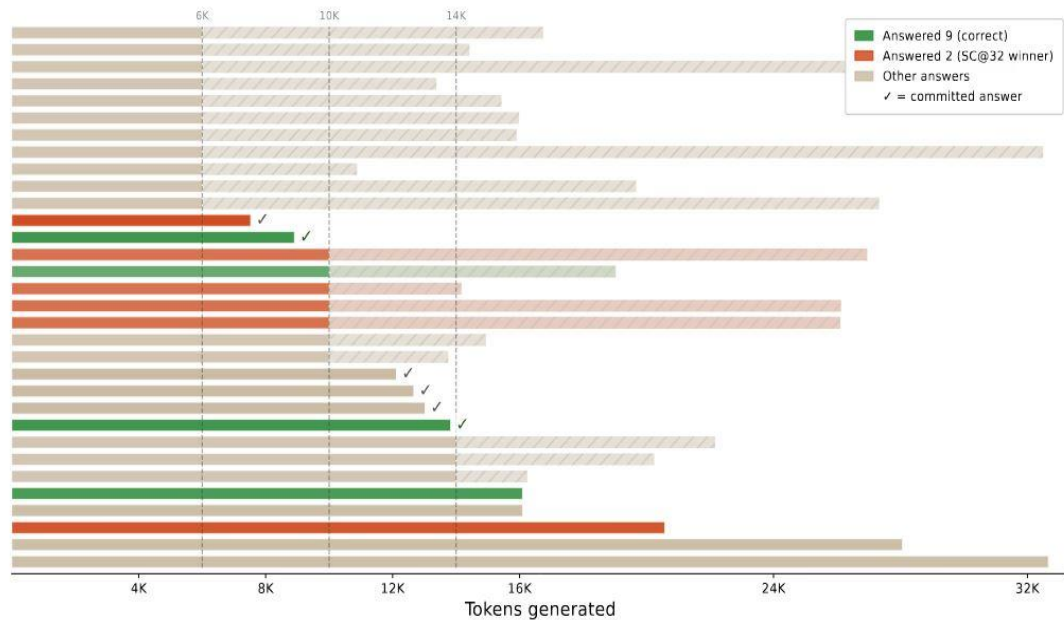
Results

- Small open-source model + RAG approaches GPT-4
- Medical-specialized models (Meerkat · OpenBioLLM) rank in the bottom 4 of 12

Takeaway

Benchmark scores and clinical usefulness do not point the same way
An LLM's knowledge does not necessarily translate into generation quality

Funnel-of-Thoughts: Efficient Test-Time Scaling via Early Voting and Rollout Pruning



Problem

Self-Consistency@32 is accurate but costs 32× the generation
Existing savings need a reward model or logit access → extra training or model privileges

Insight

Across 115,200 reasoning traces, failing rollouts show dense hesitation markers;
successful ones converge early.

Method

Early Voting: count responses that conclude early toward the final answer
Rollout Pruning: prune unproductive reasoning paths at a fixed ratio

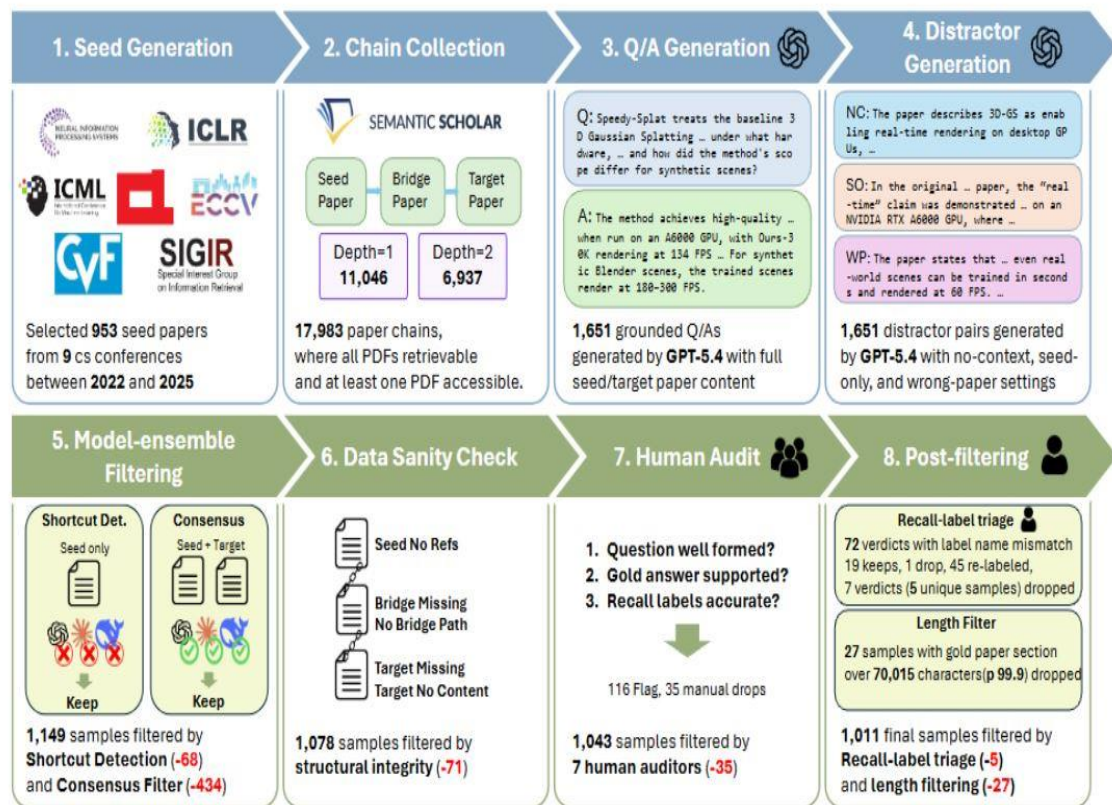
Results

56% fewer attention FLOPs, 37.6% less wall time, at SC@32 accuracy
Pruning that transfers across domains and models with no retraining, reward model, or logit access

Status

On arXiv · Korean patent filed (multi-sample inference acceleration)

AgentHop: A Diagnostic Benchmark for Agentic Multi-Hop Scientific Question Answering



Problem

When an agent fails a multi-hop scientific question, accuracy alone cannot say whether **search, synthesis, tool use, or resource management** broke

Data construction

An 8-stage generation pipeline

- ① 953 seed papers from 9 CS venues
 - ② 17,983 citation chains (depth 1–2)
 - ③ 1,651 grounded Q/A
 - ④ distractor generation
 - ⑤ model-ensemble filtering (–502)
 - ⑥ structural checks (–71)
 - ⑦ expert audit (–35)
 - ⑧ post-filtering (–32)
- final benchmark of **1,011** items

Four diagnostic axes

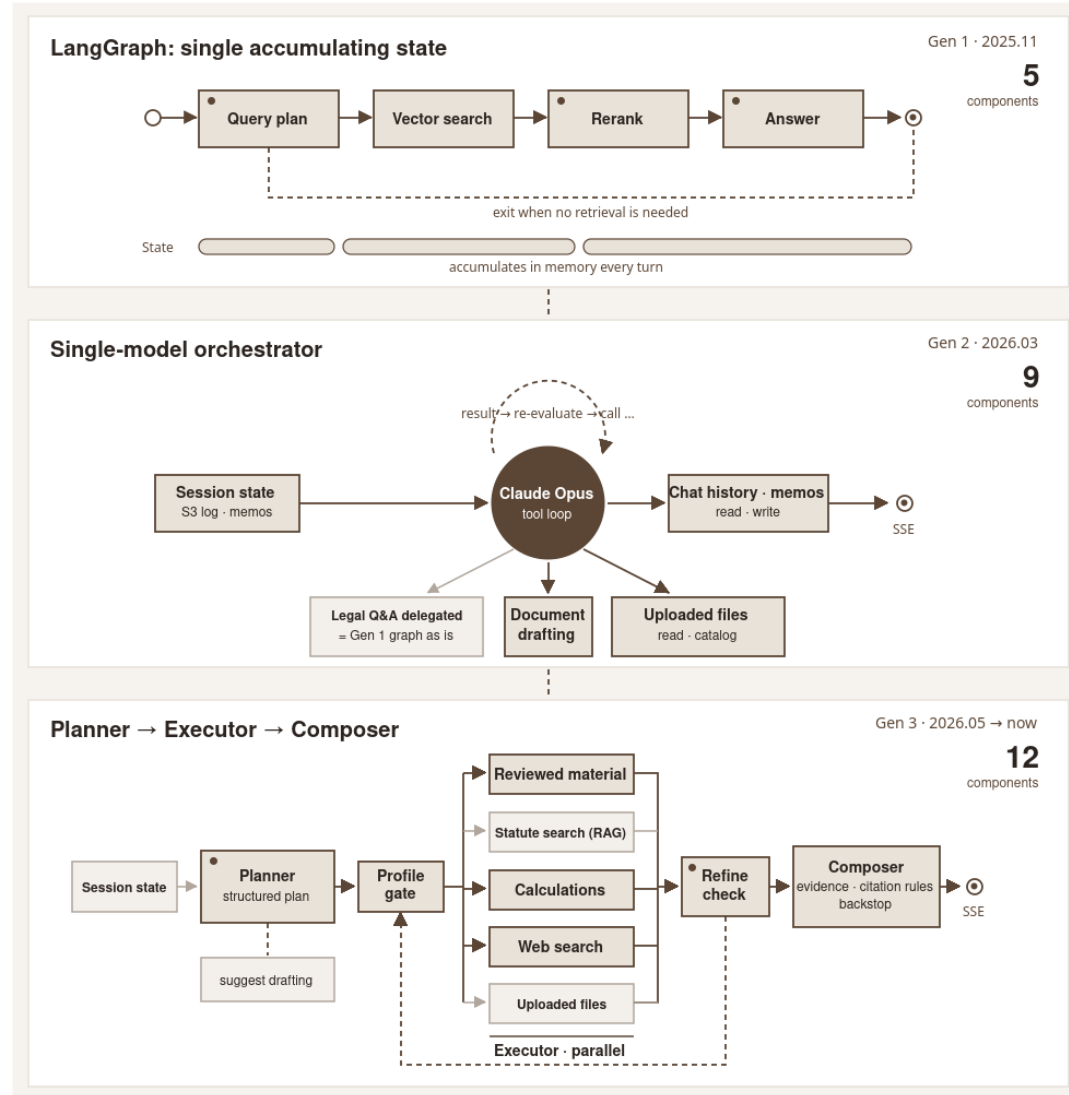
1. Search – recall-based agentic search
2. Synthesis – evidence converted into answers
3. Tool-use – calls per turn and usage patterns
4. Resource – token, turn, and budget overruns

Results & status

1. Agentic tasks involve far more, and more varied, failure points than conventional tasks
2. The model that searches best (Opus 4.6) is not the one that writes answers best (Gemini-3 Pro)
3. At the same size, newer agent-trained generations (Qwen3 → Qwen 3.5) improve consistently across all four axes

Accepted at NeurIPS 2026 Evaluations & Datasets

아는노무사: A Legal-RAG Agent for Labor Lawyers



Role · Team

- AI tech lead on a 4-person team (2 AI · 1 full-stack · 1 practicing labor attorney)
- **RAG**: retrieval pool from 10 public legal databases (statutes, precedents), web search, and attorney-reviewed material
- **Chat agent**: legal consultation, review, document drafting and editing through a single entry point

Three agent architectures

- LangGraph single-state design: per-turn state growth and rigidity
- Single large-model orchestrator: cost grew with every added feature
- Planner (structured plan) → Executor (parallel execution) → Composer (generation)
- Parallelized and serialized stages to cut time-to-first-token
- Separating planning from generation improved both cost and quality
- Time to first token: 2 min 30 s → 20–30 s

Infrastructure

Self-hosted embedding service

Session context offloaded to AWS S3

Per-request cost tracing with LangTrace

Internal evaluation benchmark and a debug portal for continuous quality control

Status

1,100+ visitors during the July closed beta; PoC complete, preparing for launch